In [14]:
```
import pandas as pd

# Create a dummy DataFrame with 'Weight', 'Volume', and 'CO2' columns
dummy_data = {
    'Weight': [1000, 1200, 1500, 1100, 1300, 1400, 1600, 900, 1050, 1250],
    'Volume': [15, 18, 22, 16, 20, 21, 25, 14, 17, 19],
    'CO2': [99, 105, 115, 102, 110, 112, 120, 95, 100, 108]
}
df_dummy = pd.DataFrame(dummy_data)

# Save the DataFrame to a CSV file named 'data.csv'
df_dummy.to_csv('data.csv', index=False)

print("Created 'data.csv' with dummy data.")
```

```
Created 'data.csv' with dummy data.
```

In [15]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor
# STEP 1: Load data
df = pd.read_csv("data.csv")
print(df.head())
print(df.shape)
print(df.info())
# STEP 2: Define independent and dependent variables
X = df[['Weight', 'Volume']]
y = df['CO2']
# STEP 3: Create and fit Linear Regression model
model = LinearRegression()
model.fit(X, y)
# STEP 4: Display regression coefficients
print("Intercept:", model.intercept_)
print("Coefficients:", model.coef_)
# STEP 5: Make predictions
y_pred = model.predict(X)
print("\nPredicted CO2 values:")
print(y_pred)
# STEP 6: Model evaluation
r2 = r2_score(y, y_pred)
mse = mean_squared_error(y, y_pred)
rmse = np.sqrt(mse)
print("\nR-squared:", r2)
print("MSE:", mse)
print("RMSE:", rmse)
# STEP 7: Calculate residuals
residuals = y - y_pred
print("\nResiduals:")
print(residuals)
# STEP 8: Residual Plot
plt.figure(figsize=(8, 5))
```

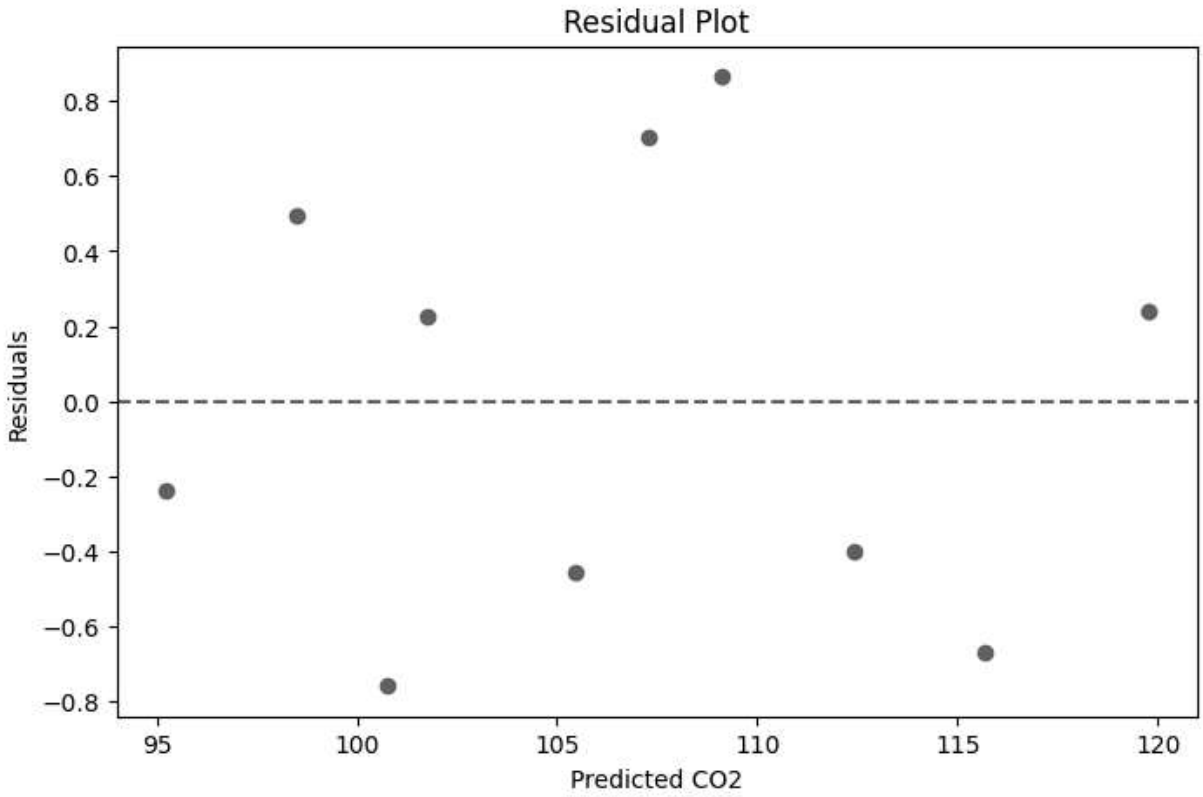
```
plt.scatter(y_pred, residuals)
plt.axhline(y=0, linestyle='--')
plt.xlabel("Predicted CO2")
plt.ylabel("Residuals")
plt.title("Residual Plot")
plt.show()
# STEP 9: Calculate VIF
X_vif = sm.add_constant(X)
vif = pd.DataFrame()
vif["Variable"] = X_vif.columns
vif["VIF"] = [
    variance_inflation_factor(X_vif.values, i)
    for i in range(X_vif.shape[1])
]
print("\nVIF:")
print(vif)
# STEP 10: Fit OLS model
X_sm = sm.add_constant(X)
model_sm = sm.OLS(y, X_sm).fit()
print("\nOLS Regression Summary:")
print(model_sm.summary())
# STEP 11: Calculate Cook's Distance
influence = model_sm.get_influence()
cooks_distance = influence.cooks_distance[0]
print("\nCook's Distance:")
for i, value in enumerate(cooks_distance):
    print("Observation", i + 1, ":", value)
# STEP 12: Plot Cook's Distance
plt.figure(figsize=(10, 5))
plt.stem(
    range(1, len(cooks_distance) + 1),
    cooks_distance
)
plt.xlabel("Observation")
plt.ylabel("Cook's Distance")
plt.title("Cook's Distance")
plt.show()
```

```
   Weight  Volume  CO2
0    1000      15   99
1    1200      18  105
2    1500      22  115
3    1100      16  102
4    1300      20  110
(10, 3)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Weight  10 non-null     int64
 1   Volume  10 non-null     int64
 2   CO2     10 non-null     int64
dtypes: int64(3)
memory usage: 372.0 bytes
None
Intercept: 63.768949102892684
Coefficients: [0.02855364 0.41230319]

Predicted CO2 values:
[ 98.50714024 105.45477847 115.67008422 101.77480776 109.13474918
 112.4024167  119.76235811  95.23947272 100.75942878 107.29476382]

R-squared: 0.9944823175644987
MSE: 0.304796777737094
RMSE: 0.5520840314092539

Residuals:
0    0.492860
1   -0.454778
2   -0.670084
3    0.225192
4    0.865251
5   -0.402417
6    0.237642
7   -0.239473
8   -0.759429
9    0.705236
Name: CO2, dtype: float64
```

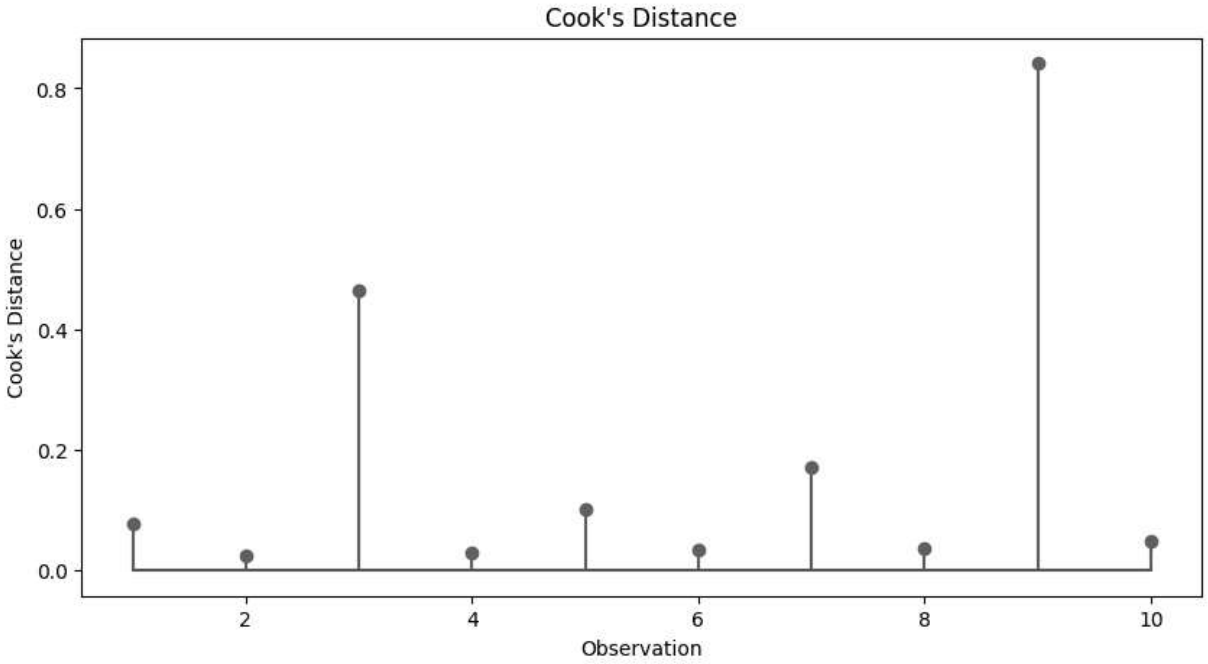

Residual Plot
Residuals
Predicted CO2
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
95
100
105
110
115
120

```
VIF:
  Variable        VIF
0    const  34.672281
1   Weight  34.763530
2   Volume  34.763530

OLS Regression Summary:
                            OLS Regression Results
================================================================================
Dep. Variable:                    CO2   R-squared:                       0.994
Model:                            OLS   Adj. R-squared:                  0.993
Method:                 Least Squares   F-statistic:                     630.8
Date:                Tue, 08 Sep 2026   Prob (F-statistic):           1.25e-08
Time:                        00:57:53   Log-Likelihood:                -8.2488
No. Observations:                  10   AIC:                             22.50
Df Residuals:                       7   BIC:                             23.41
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         63.7689      1.229     51.899      0.000      60.864      66.674
Weight         0.0286      0.006      4.956      0.002       0.015       0.042
Volume         0.4123      0.381      1.081      0.315      -0.489       1.314
==============================================================================
Omnibus:                        2.086   Durbin-Watson:                   2.236
Prob(Omnibus):                  0.352   Jarque-Bera (JB):                0.857
Skew:                           0.137   Prob(JB):                        0.651
Kurtosis:                       1.592   Cond. No.                     7.36e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly spe
cified.
[2] The condition number is large, 7.36e+03. This might indicate that there are
strong multicollinearity or other numerical problems.

Cook's Distance:
Observation 1 : 0.07795442024150274
Observation 2 : 0.02542550070007861
Observation 3 : 0.46343595530675885
Observation 4 : 0.028903641664023882
Observation 5 : 0.10132471385579798
Observation 6 : 0.03351982164282909
Observation 7 : 0.17214081356915154
Observation 8 : 0.037258856579555126
Observation 9 : 0.8414608312899107
Observation 10 : 0.04751152884924505
```



In [ ]:
```
!jupyter nbconvert --to html /content/smld_5_new.ipynb
```

```
[NbConvertApp] Converting notebook /content/smld_5_new.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 1 image(s).
[NbConvertApp] Writing 317999 bytes to /content/smld_5_new.html
```